LAB 3&4

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B-11

24. WRITE A C PROGRAM TO FIND THE VALUE OF Y USING

Y(x,n) = 1+x where n=1 1+x/n where n=2

1 +xn when n=3 1+ nx when n>3 or n<1

Program_24.c > main()

```c
#include <stdio.h>
int main()
{
    int x, y, n;
    printf("Enter the value of n : ");
    scanf("%d", &n);
    printf("Enter the value of x : ");
    scanf("%d", &x);
    if(n==1){
        y = 1 + x;
    }
    else if(n==2){
        y = 1 + (x/n);
    }
    else if(n==3){
        y = 1 + (x*n);
    }
    else{
        y = 1 + (n*x);
    }
    printf("The value of y(%d, %d) = %d\n", x, n, y);
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter the value of n : 2
Enter the value of x : 2
The value of y(2, 2) = 2
PS C:\Users\Lenovo\Desktop\C>
```

25. PROGRAM TO PRINT TABLE OF ANY NO.

Program_25.c > main()

```c
#include <stdio.h>
int main()
{
    int num, i;

    printf("Enter the num : ");
    scanf("%d", &num);

    for(i=1; i<11; i++)
    {
        printf("%d\n", num*i);
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ;
Enter the num : 4
4
8
12
16
20
24
28
32
36
40
PS C:\Users\Lenovo\Desktop\C>
```

26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.

Program_26.c > main()

```c
#include<stdio.h>
int main()
{
    int i, n, even=0, odd=0;

    printf("Enter the value of n : ");
    scanf("%d", &n);

    for(i=1; i<=n; i++)
    {
        if(i % 2 == 0)
        {
            even = even + i;
        }
        else
        {
            odd = odd + i;
        }
    }

    printf("Sum of even numbers = %d\n", even);
    printf("Sum of odd numbers = %d", odd);

    return 0;
}
```

Problems Output Debug Console **Terminal** Ports

```
Enter the value of n : 10
Sum of even numbers = 30
Sum of odd numbers = 25
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if
Enter the value of n : 5
Sum of even numbers = 6
Sum of odd numbers = 9
PS C:\Users\Lenovo\Desktop\C>
```

27. PROGRAM TO CHECK WHETHER A NO IS PRIME OR NOT.

Program_27.c > main()

```c
#include <stdio.h>
int main(){
    int n, i, factor = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n <= 1){
        printf("Not a prime number");
    }
    else{
        for (i = 1; i <= n; i++)
        {
            if (n % i == 0){
                factor++;
            }
        }
        if (factor == 2){
            printf("Number is a Prime number");
        }
        else{
            printf("Number is not a Prime number");
        }
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) {
Enter a number: 17
Number is a Prime number
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) {
Enter a number: 8
Number is not a Prime number
PS C:\Users\Lenovo\Desktop\C>
```

28. PROGRAM TO CHECK WHETHER A NO IS ARMSTRONG OR NOT.

Program_28.c > main()

```c
#include <stdio.h>
int main()
{
    int num, q, sum=0, tamp=0;

    printf("Enter num : ");
    scanf("%d", &num);
    tamp=num;

    while(num != 0)
    {
        q = num % 10;
        sum = sum+q*q*q;
        num = num/10;
    }
    if(tamp==sum)
    {
        printf("Num is Armstrong");
    }
    else
    {
        printf("Num is not Armstrong");
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { g
Enter num : 153
Num is Armstrong
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { g
Enter num : 23
Num is not Armstrong
```

29. PROGRAM TO CHECK WHETHER A NO IS PALINDROM OR NOT.

Program_29.c > main()

```c
#include <stdio.h>
int main()
{
    int num, q, sum=0, tamp=0;

    printf("Enter num : ");
    scanf("%d", &num);
    tamp=num;

    while(num != 0)
    {
        q = num%10;
        sum = sum*10+q;
        num = num/10;
    }
    if(tamp==sum)
    {
        printf("Num is palindrome");
    }
    else
    {
        printf("Num is not palindrome");
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter num : 1221
Num is palindrome
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter num : 1234
Num is not palindrome
PS C:\Users\Lenovo\Desktop\C>
```

30. PROGRAM TO REVERSE OF A GIVEN NO.

Program_30.c > main()

```c
#include <stdio.h>
int main()
{
    int num, q, sum=0;

    printf("Enter the value of num = ");
    scanf("%d", &num);

    while(num!=0)
    {
        q = num%10;
        sum = sum * 10 + q;
        num = num/10;
    }

    printf("Reverse number is = %d", sum);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter the value of num = 167
Reverse number is = 761
PS C:\Users\Lenovo\Desktop\C>
```

31. PROGRAM TO PRINT SUM OF INDIVIDUAL DIGITS OF N DIGIT NO.

Program_31.c > main()

```c
#include <stdio.h>
int main()
{
    int num, q, sum=0;

    printf("Enter the value of num = ");
    scanf("%d", &num);

    while(num!=0)
    {
       q = num%10;
       sum = sum + q;
       num = num/10;
    }

    printf("sum of digits = %d", sum);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if
Enter the value of num = 1234
sum of digits = 10
PS C:\Users\Lenovo\Desktop\C>
```

32. PROGRAM TO FINDOUT MAX AND SECOND MAX FROM N NOS.

Program_32.c > main()

```c
#include <stdio.h>
int main(){
    int n, num, fmax, smax, i;
    printf("Enter how many numbers : ");
    scanf("%d", &n);
    printf("Enter number : ");
    scanf("%d", &num);
    fmax=num;
    smax=num;
    for(i=2; i<=n; i++){
        printf("Enter number ");
        scanf("%d", &num);
        if(num>fmax)
        {
            smax=fmax;
            fmax=num;
        }
        if(num<fmax && num>smax)
        {
            smax=num;
        }
    }
    printf("Maximum number = %d\n", fmax);
    printf("Second maximum number = %d", smax);
    return 0;
```

Problems Output Debug Console Terminal Ports

```
Enter how many numbers : 5
Enter number : 1
Enter number 4
Enter number 67
Enter number 7
Enter number 36
Maximum number = 67
Second maximum number = 36
```

33. PROGRAM TO FINDOUT MAX AND MIN FROM N NOS.

C Program_33.c > main()

```c
#include <stdio.h>
int main(){
    int num, max, min, i, n;
    printf("Enter how many numbers : ");
    scanf("%d",&n);
    printf("Enter num : ");
    scanf("%d",&num);
    max=num;
    min=num;
    for(i=1; i<=n-1; i++){
        printf("Enter num : ");
        scanf("%d",&num);
        if(max<num)
        {
            max=num;
        }
        if(min>num)
        {
            min=num;
        }
    }
    printf("max number is = %d\n", max);
    printf("min number is = %d\n", min);
    return 0;
}
```

Problems Output Debug Console **Terminal** Ports

```
Enter how many numbers : 5
Enter num : 4
Enter num : 77
Enter num : 8
Enter num : 36
Enter num : 45
max number is = 77
min number is = 4
PS C:\Users\Lenovo\Desktop\C>
```

34. PROGRAM TO PRINT FIBONACCI SERIES.(0,1,1,2,3 ...)

Program_34.c > main()

```c
#include<stdio.h>
int main()
{
    int a=0, b=1, c, num;

    printf("Enter num : ");
    scanf("%d", &num);

    while(num!=0)
    {
        c = a + b;
        b = a;
        printf("%d\t", b);
        a = c;
        num--;
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc Program_34.c
Enter num : 10
0       1       1       2       3       5       8       13      21      34
PS C:\Users\Lenovo\Desktop\C>
```

35. READ A NUMBER N AND PRINT A SINGLE DIGIT ANSWER SHOWING SUM OF DIGIT OF N.

Program_35.c > main()

```c
#include<stdio.h>
int main()
{
    int num, sum;

    printf("Enter a number : ");
    scanf("%d", &num);

    while(num>=10)
    {
        sum=0;

        while(num>0)
        {
            sum=sum + num%10;
            num = num/10;
        }

        num=sum;
    }

    printf("Single digit answer = %d", num);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter a number : 12345
Single digit answer = 6
PS C:\Users\Lenovo\Desktop\C>
```

36. PROGRAM TO PRINT 1,3,5,7,9

```c
{
    int n, i,j;

    printf("Enter n : ");
    scanf("%d", &n);

    for(i=1; i<=n; i=i+2)
    {
        printf("%d\t", i);
    }

    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc Progr
Enter n : 10
1       3       5       7       9
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc Progr
Enter n : 20
1       3       5       7       9       11      13      15      17      19
PS C:\Users\Lenovo\Desktop\C>
```

37. PROGRAM TO PRINT 2,4,6,8,10,12 ..... N.

```c
#include<stdio.h>
int main()
{
    int n, i;

    printf("Enter n : ");
    scanf("%d", &n);

    for(i=0; i<=n; i=i+2)
    {
        printf("%d\t", i);
    }

    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc Program_37.c
Enter n : 10
0       2       4       6       8       10
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc Program_37.c
Enter n : 20
0       2       4       6       8       10      12      14      16      18      20
PS C:\Users\Lenovo\Desktop\C>
```

38. PROGRAM TO PRINT 1,4,9,16,25, ... N.

```c
#include <stdio.h>
int main()
{
    int n, i;

    printf("Enter n : ");
    scanf("%d", &n);

    for(i=1; i<=n; i++)
    {
        printf("%d\t", i*i);
    }

    return 0;
}
```

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc Program_
Enter n : 10
1       4       9       16      25      36      49      64      81      100
PS C:\Users\Lenovo\Desktop\C>
```

39. PROGRAM TO PRINT 1/1!+2/2!+3/3!+4/4!, ... N.

C Program_39.c > main()

```c
#include<stdio.h>
int main()
{
    int n, i, j;
    float fact, sum=0;

    printf("Enter the value of n : ");
    scanf("%d", &n);

    for(i=1;i<=n;i++)
    {
        fact=1;

        for(j=1;j<=i;j++)
        {
            fact=fact*j;
        }

        sum=sum + i/fact;
    }

    printf("Sum of the series = %f", sum);

    return 0;
}
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if
Enter the value of n : 4
Sum of the series = 2.666667
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if
Enter the value of n : 3
Sum of the series = 2.500000
PS C:\Users\Lenovo\Desktop\C>
```

40. READ NUMBERS TILL A NEGATIVE NUMBER IS ENTERED AND CALCULATE SUM OF A LIST OF NUMBERS READ.

C Program_40.c > main()

```c
#include<stdio.h>
int main()
{
    int num, sum=0;

    printf("Enter numbers : \n");
    scanf("%d", &num);

    while(num>=0)
    {
        sum=sum+num;
        scanf("%d", &num);
    }

    printf("Sum of numbers = %d", sum);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter numbers :
12
8
10
5
-12334
Sum of numbers = 35
PS C:\Users\Lenovo\Desktop\C>
```

41. READ A NUMBER N AND PRINT FACTOR OF N.(EX 28= 1,2,4,7,14,28)

Program_41.c > main()

```c
#include <stdio.h>
int main()
{
    int n, i;

    printf("Enter a number : ");
    scanf("%d", &n);

    printf("Factors of given number are : ");
    for(i=1; i<=n; i++)
    {
        if(n % i == 0 )
        {
            printf("%d\t", i);
        }
    }

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc
Enter a number : 28
Factors of given number are : 1 2       4       7       14      28
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc
Enter a number : 12
Factors of given number are : 1 2       3       4       6       12
PS C:\Users\Lenovo\Desktop\C>
```

42. READ A NUMBER N AND PRINT IF N IS PRIME OR COMPOSITE.

C Program_42.c > main()

```c
#include <stdio.h>
int main()
{
    int n, i, factor = 0;
    printf("Enter a number: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        if (n % i == 0)
        {
            factor++;
        }
    }

    if (factor == 2)
    {
        printf("Number is Prime number");
    }
    else
    {
        printf("Number is Composite number");
    }
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if (
Enter a number: 17
Number is Prime number
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if (
Enter a number: 12
Number is Composite number
PS C:\Users\Lenovo\Desktop\C>
```

43. WRITE A C PROGRAM TO CALCULATE THE COMMISSION FOR A SALES REPRESENTATIVE AS PER THE SALES AMOUNT GIVEN BELOW.
If sales <= Rs. 500, commission is 5%
If sales >500 but <= 2000, commission is Rs. 35 plus 10% above Rs. 500
If saleS >2000 but <= 5000, commission is Rs. 185 plus 12% above Rs.
2000
If saleS >5000, commission is 12.5%

Program_43.c > main()

```c
int main()
{
    int sales, commission;
    printf("Enter the value of sales : ");
    scanf("%d", &sales);
    if(sales<=500){
        commission = sales*(0.05);
        printf("Commission is = %d", commission);
    }
    if(sales>500 && sales<=2000){
        commission = 35 + (sales-500)*(0.1);
        printf("Commission is = %d", commission);
    }
    if(sales>2000 && sales<=5000){
        commission = 185 + (sales-2000)*(0.12);
        printf("Commission is = %d", commission);
    }
    if(sales>5000){
        commission = sales*(0.125);
        printf("Commission is = %d", commission);
    }
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gc
Enter the value of sales : 600
Commission is = 45
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gc
Enter the value of sales : 6000
Commission is = 750
PS C:\Users\Lenovo\Desktop\C>
```

44. WRITE A PROGRAM IN C FOR THE FOLLOWING.
AN ELECTRIC DISTRIBUTION COMPANY CHARGES ITS DOMESTIC CONSUMERS AS FOLLOWS.

Consumption in unit	Rate for Charge
0-200	Re 0.50 per unit
210-400	Rs. 100 plus Re 0.65 per unit excess of 200
401-600	Rs. 230 plus Re 0.80 per unit excess of 400
Above 600	Rs. 425 plus Rs. 125 per unit excess of 600

Print the amount to be paid by the consumer.

Program_44.c > main()

```c
int main()
{
    int unit, rate;
    printf("Enter the value of consumption in unit : ");
    scanf("%d", &unit);
    if(unit>=0 && unit<=200){
        rate =  (0.5)*unit;
        printf("Rate for charge = %d", rate);
    }
    if(unit>=201 && unit<=400){
        rate = 100 + 0.65*(unit-200);
        printf("Rate for charge = %d", rate);
    }
    if(unit>=401 && unit<=600){
        rate = 230 + 0.8*(unit-400);
        printf("Rate for charge = %d", rate);
    }
    if(unit>600){
        rate = 425 + 125*(unit-600);
        printf("Rate for charge = %d", rate);
    }
    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) {
Enter the value of consumption in unit : 100
Rate for charge = 50
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) {
Enter the value of consumption in unit : 601
Rate for charge = 550
PS C:\Users\Lenovo\Desktop\C>
```